

SVM Example

SVM Primal (hard margin)

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to $y_i(w \cdot x_i + b) \geq 1$

SVM Dual

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to $\sum_{i=1}^n \alpha_i y_i = 0, \quad \alpha_i \geq 0$

Dataset

Class +1:

- $x_1 = (2, 2), y_1 = +1$
- $x_2 = (4, 4), y_2 = +1$

Class -1:

- $x_3 = (2, 0), y_3 = -1$
- $x_4 = (0, 2), y_4 = -1$
- $x_5 = (3, 1), y_5 = -1$

Dual Form

$$\max_{\alpha} \sum_{i=1}^5 \alpha_i - \frac{1}{2} \sum_{i=1}^5 \sum_{j=1}^5 \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to:

$$\sum_{i=1}^5 \alpha_i y_i = 0, \quad \alpha_i \geq 0$$

Compute Dot Products

ij	Value
$x_1 \cdot x_1$	8
$x_1 \cdot x_2$	16
$x_1 \cdot x_3$	4
$x_1 \cdot x_4$	4
$x_1 \cdot x_5$	8
$x_2 \cdot x_2$	32
$x_2 \cdot x_3$	8
$x_2 \cdot x_4$	8
$x_2 \cdot x_5$	16
$x_3 \cdot x_3$	4
$x_3 \cdot x_4$	0
$x_3 \cdot x_5$	6
$x_4 \cdot x_4$	4
$x_4 \cdot x_5$	2
$x_5 \cdot x_5$	10

$$\max_{\alpha} \sum_{i=1}^5 \alpha_i - \frac{1}{2} \sum_{i=1}^5 \sum_{j=1}^5 \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to:

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Support Vector $y_i(w \cdot x_i + b) = 1$

Assume Boundary eqn: $x_1 + x_2 - 3 = 0$

For $x_1 = (2, 2)$:

$$2 + 2 - 3 = 1 \Rightarrow y(w \cdot x + b) = 1$$

For $x_3 = (2, 0)$:

$$2 + 0 - 3 = -1 \Rightarrow y(w \cdot x + b) = (-1)(-1) = 1$$

For $x_4 = (0, 2)$:

$$0 + 2 - 3 = -1 \Rightarrow y(w \cdot x + b) = 1$$

Evaluate for $x_5 = (3, 1)$:

$$3 + 1 - 3 = 1$$

Then:

$$y_5(w \cdot x_5 + b) = (-1)(1) = -1$$

This is **far from margin**, not equal to 1

KKT(Karush–Kuhn–Tucker) condition

KKT condition says:

KKT conditions are a set of **necessary conditions for optimality** in a constrained optimization problem.

In simple words: They tell us **when a solution is optimal** for problems with constraints.

Either a constraint is active **OR** its multiplier is zero — never both inactive.

From optimization (KKT condition):

In SVM Primal

Constraint:

$$y_i(w \cdot x_i + b) \geq 1$$

Rewrite:

$$g_i = 1 - y_i(w \cdot x_i + b) \leq 0$$

Case 1:

$$\alpha_i(y_i(w \cdot x_i + b) - 1) = 0$$

$$\alpha_i > 0 \Rightarrow y_i(w \cdot x_i + b) = 1$$

Point is **on boundary** (support vector)

Case 2:

$$y_i(w \cdot x_i + b) > 1 \Rightarrow \alpha_i = 0$$

Point is not important

If point is on boundary, why $\alpha > 0$

$$y_i(w \cdot x_i + b) = 1 \quad (\text{boundary})$$

$$(y_i(w \cdot x_i + b) - 1) = 0$$

$$\alpha_i \cdot 0 = 0$$

If point is **NOT** on boundary, why $\alpha = 0$

$$y_i(w \cdot x_i + b) > 1$$

$$(y_i(w \cdot x_i + b) - 1) > 0$$

$$\alpha_i \cdot (\text{positive}) = 0$$

$$\alpha_i = 0$$

Point position	Condition	Result
On margin	(= 1)	($\alpha > 0$) possible
Outside margin	(> 1)	($\alpha = 0$)
Inside margin (soft)	(< 1)	($\alpha > 0$)

Assume Support Vectors

- $x_1 = (2, 2), y = +1$
- $x_3 = (2, 0), y = -1$
- $x_4 = (0, 2), y = -1$

So:

$$\alpha_2 = 0, \quad \alpha_5 = 0$$

Constraint

$$\max_{\alpha} \sum_{i=1}^5 \alpha_i - \frac{1}{2} \sum_{i=1}^5 \sum_{j=1}^5 \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to:

$$\sum_{i=1}^5 \alpha_i y_i = 0, \quad \alpha_i \geq 0$$

$$\alpha_1(1) + \alpha_3(-1) + \alpha_4(-1) = 0$$

$$\alpha_1 = \alpha_3 + \alpha_4$$

Dual Objective (Only 3 Variables)

$$\sum_{i=1}^5 \alpha_i - \frac{1}{2} \sum_{i=1}^5 \sum_{j=1}^5 \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

$$L = \alpha_1 + \alpha_3 + \alpha_4 - \frac{1}{2} Q$$

$$Q = \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

Now compute Q :

$$Q = \alpha_1^2(8) + \alpha_3^2(4) + \alpha_4^2(4)$$

Cross terms:

$$+2\alpha_1\alpha_3(1)(-1)(4) + 2\alpha_1\alpha_4(1)(-1)(4) + 2\alpha_3\alpha_4(-1)(-1)(0)$$

Simplify:

$$Q = 8\alpha_1^2 + 4\alpha_3^2 + 4\alpha_4^2 - 8\alpha_1\alpha_3 - 8\alpha_1\alpha_4$$

$$Q = \sum_{i=1,3,4} \sum_{j=1,3,4} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

$$\begin{aligned} Q = & \alpha_1 \alpha_1 y_1 y_1 (x_1 \cdot x_1) \\ & + \alpha_1 \alpha_3 y_1 y_3 (x_1 \cdot x_3) \\ & + \alpha_1 \alpha_4 y_1 y_4 (x_1 \cdot x_4) \\ & + \alpha_3 \alpha_1 y_3 y_1 (x_3 \cdot x_1) \\ & + \alpha_3 \alpha_3 y_3 y_3 (x_3 \cdot x_3) \\ & + \alpha_3 \alpha_4 y_3 y_4 (x_3 \cdot x_4) \\ & + \alpha_4 \alpha_1 y_4 y_1 (x_4 \cdot x_1) \\ & + \alpha_4 \alpha_3 y_4 y_3 (x_4 \cdot x_3) \\ & + \alpha_4 \alpha_4 y_4 y_4 (x_4 \cdot x_4) \end{aligned}$$

Term (1,1)

Substitute:

- $\alpha_1 \alpha_1 = \alpha_1^2$
- $y_1 = +1 \Rightarrow y_1 y_1 = 1$
- $x_1 = (2, 2) \Rightarrow x_1 \cdot x_1 = 4 + 4 = 8$

$$= 8\alpha_1^2$$

Term (3,3)

- $y_3 = -1 \Rightarrow y_3^2 = 1$
- $x_3 = (2, 0) \Rightarrow 2^2 + 0^2 = 4$

$$\alpha_3^2 y_3^2 (x_3 \cdot x_3)$$

$$= 4\alpha_3^2$$

Term (4,4)

- $y_4 = -1 \Rightarrow y_4^2 = 1$
- $x_4 = (0, 2) \Rightarrow 0 + 4 = 4$

$$\alpha_4^2 y_4^2 (x_4 \cdot x_4)$$

$$= 4\alpha_4^2$$

$$\alpha_1 \alpha_1 y_1 y_1 (x_1 \cdot x_1)$$

◆ (1,3)

- $y_1 y_3 = (1)(-1) = -1$
- $x_1 \cdot x_3 = 2 \cdot 2 + 2 \cdot 0 = 4$

$$\alpha_1 \alpha_3 y_1 y_3 (x_1 \cdot x_3)$$

$$= -4\alpha_1 \alpha_3$$

◆ (3,1)

$$\alpha_3 \alpha_1 y_3 y_1 (x_3 \cdot x_1)$$

$$= -4\alpha_1 \alpha_3$$

Same values:

◆ (1,4)

- $y_1 y_4 = (1)(-1) = -1$
- $x_1 \cdot x_4 = 2 \cdot 0 + 2 \cdot 2 = 4$

$$\alpha_1 \alpha_4 y_1 y_4 (x_1 \cdot x_4)$$

$$= -4\alpha_1 \alpha_4$$

◆ (4,1)

Same:

$$= -4\alpha_1 \alpha_4$$

◆ (3,4)

- $y_3 y_4 = (-1)(-1) = 1$
- $x_3 \cdot x_4 = 2 \cdot 0 + 0 \cdot 2 = 0$

$$\alpha_3 \alpha_4 y_3 y_4 (x_3 \cdot x_4)$$

$$= 0$$

◆ (4,3)

Same:

$$= 0$$

$$(1, 3) = -4\alpha_1 \alpha_3$$

$$(3, 1) = -4\alpha_1 \alpha_3$$

$$(1, 4) = -4\alpha_1 \alpha_4$$

$$(4, 1) = -4\alpha_1 \alpha_4$$

$$(3, 4) = 0$$

$$(4, 3) = 0$$

$$Q_{\text{cross}} = -8\alpha_1 \alpha_3 - 8\alpha_1 \alpha_4 + 0$$

Substitute Constraint $Q = 8\alpha_1^2 + 4\alpha_3^2 + 4\alpha_4^2 - 8\alpha_1\alpha_3 - 8\alpha_1\alpha_4$

$$\alpha_1 = \alpha_3 + \alpha_4$$

Substitute into Q :

$$Q = 8(\alpha_3 + \alpha_4)^2 + 4\alpha_3^2 + 4\alpha_4^2 - 8(\alpha_3 + \alpha_4)\alpha_3 - 8(\alpha_3 + \alpha_4)\alpha_4$$

Expand:

$$Q = 8(\alpha_3^2 + 2\alpha_3\alpha_4 + \alpha_4^2) + 4\alpha_3^2 + 4\alpha_4^2 - 8(\alpha_3^2 + \alpha_3\alpha_4) - 8(\alpha_3\alpha_4 + \alpha_4^2)$$

Simplify:

$$Q = (8 + 4 - 8)\alpha_3^2 + (16 - 8 - 8)\alpha_3\alpha_4 + (8 + 4 - 8)\alpha_4^2$$

$$Q = 4\alpha_3^2 + 0 + 4\alpha_4^2$$

Final Dual Objective

$$L = \alpha_1 + \alpha_3 + \alpha_4 - \frac{1}{2}Q$$

$$L = (\alpha_3 + \alpha_4) + \alpha_3 + \alpha_4 - \frac{1}{2}(4\alpha_3^2 + 4\alpha_4^2)$$

$$L = 2\alpha_3 + 2\alpha_4 - 2\alpha_3^2 - 2\alpha_4^2$$

$$\alpha_1 = \alpha_3 + \alpha_4$$

$$Q = 4\alpha_3^2 + 0 + 4\alpha_4^2$$

Maximize

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

$$\text{subject to } \sum_{i=1}^n \alpha_i y_i = 0, \quad \alpha_i \geq 0$$

$$L = 2\alpha_3 + 2\alpha_4 - 2\alpha_3^2 - 2\alpha_4^2$$

Take derivatives:

$$\frac{\partial L}{\partial \alpha_3} = 2 - 4\alpha_3 = 0 \Rightarrow \alpha_3 = 0.5$$

$$\frac{\partial L}{\partial \alpha_4} = 2 - 4\alpha_4 = 0 \Rightarrow \alpha_4 = 0.5$$

Then:

$$\alpha_1 = \alpha_3 + \alpha_4 = 1$$

Final Alphas

$$\alpha_1 = 1$$

$$\alpha_3 = 0.5$$

$$\alpha_4 = 0.5$$

$$\alpha_2 = 0, \alpha_5 = 0$$

Exactly 3 support vectors

Compute w

$$w = \sum \alpha_i y_i x_i$$

$$w = 1(1)(2, 2) + 0.5(-1)(2, 0) + 0.5(-1)(0, 2)$$

$$w = (2, 2) + (-1, 0) + (0, -1)$$

$$w = (1, 1)$$

$$\alpha_1 = 1$$

$$\alpha_3 = 0.5$$

$$\alpha_4 = 0.5$$

$$\alpha_2 = 0, \alpha_5 = 0$$

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Class -1:

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- $x_5 = (3, 1), y_5 = -1$

Compute b

All support vectors satisfy:

$$y_i(w \cdot x_i + b) = 1$$

$$x_3 = (2, 0), y_3 = -1$$

$$-1((1, 1) \cdot (2, 0) + b) = 1$$

$$-1(2 + b) = 1$$

$$2 + b = -1 \Rightarrow b = -3$$

$$x_4 = (0, 2), y_4 = -1$$

$$-1((1, 1) \cdot (0, 2) + b) = 1$$

$$-1(2 + b) = 1$$

$$b = -3$$

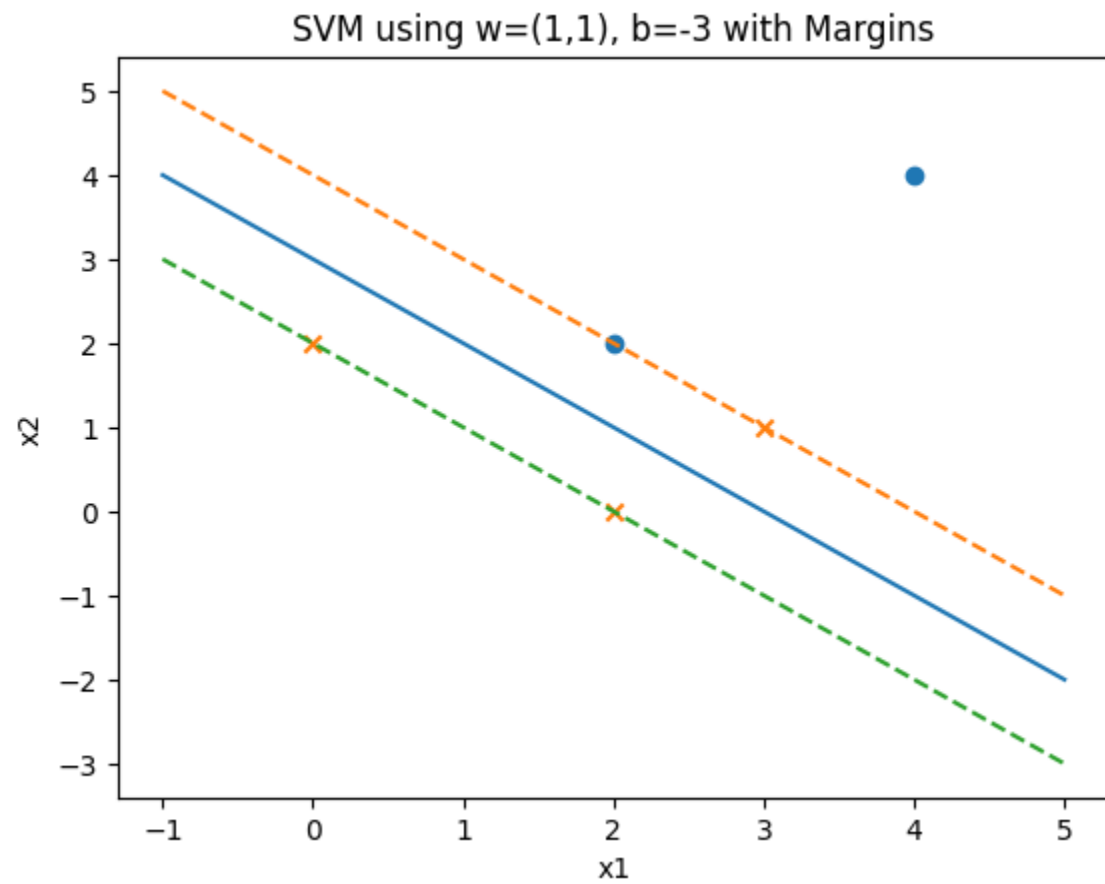
Final Step

Hyperplane:

$$w = (1, 1), \quad b = -3$$

Decision Boundary:

$$x_1 + x_2 - 3 = 0$$



Decision Boundary

$$w \cdot x + b = 0 \Rightarrow x_1 + x_2 - 3 = 0$$

$$\|w\| = \sqrt{1^2 + 1^2} = \sqrt{2} \Rightarrow \text{margin} = \frac{2}{\sqrt{2}} = \sqrt{2}$$